

Elephant Toothpaste

Science Experiment

A Hands-On Lesson for Young Scientists

Always experiment with adult supervision • Safety first!

OVERVIEW

What is Elephant Toothpaste?

Elephant Toothpaste is a spectacular and colourful science experiment that produces a large column of foam — so big it looks like it could be toothpaste for a giant elephant! It is a great way to introduce young learners to **chemical reactions**, **catalysts**, and basic **lab safety**.

Subject	Science – Chemistry
Age Group	5 – 12 years
Duration	30 – 45 minutes
Difficulty	Easy (adult supervision required)

LEARNING OBJECTIVES

By the end of this lesson, students will be able to:

- ✓ Explain what a **chemical reaction** is in simple terms.
- ✓ Describe the role of **yeast as a catalyst** in breaking down hydrogen peroxide.
- ✓ Understand why **safety equipment** (goggles & gloves) must be worn.
- ✓ Follow a multi-step experiment procedure independently.
- ✓ Connect observations to a scientific explanation.

MATERIALS NEEDED

Materials You Need

■	Hydrogen Peroxide (3 % household grade)	Available at pharmacies
■	Dish Soap	A few squirts – any brand works
■	Food Colouring	Your favourite colour!

■	Active Dry Yeast + Warm Water	1 sachet + 3 tbsp warm water
■	A Clean Bottle	Plastic or glass, narrow-neck
■	Safety Goggles & Gloves	One pair per participant
■	A Tray or Container	To catch the foam overflow

SAFETY GUIDELINES

■ ■ Safety First!

Hydrogen peroxide can irritate skin and eyes. Always handle with care. Adult supervision is **mandatory**. Do not let children handle the hydrogen peroxide directly.

- **Always ask an adult** to help with the experiment.
- **Wear safety goggles and gloves** throughout.
- Perform outdoors or on a tray — the foam can overflow significantly!
- Do **not** seal the bottle after adding the yeast mixture.
- After the experiment, dispose of materials with adult help.

STEP-BY-STEP PROCEDURE

How to Do the Experiment

Prepare the Yeast Mixture

- 1 Mix one sachet (2¼ tsp) of active dry yeast into 3 tablespoons of warm water in a small cup. Stir gently and let it sit for 2–3 minutes until frothy.

Prepare the Bottle

- 2 Place your clean, narrow-neck bottle on a tray. Ask an adult to pour about ½ cup (120 ml) of hydrogen peroxide carefully into the bottle.

Add Soap & Colour

- 3 Squirt a few drops of dish soap into the bottle. Add 5–10 drops of your favourite food colouring. Give the bottle a gentle swirl to mix — watch the colour spiral!

Pour & Watch!

- 4 Quickly pour the yeast mixture into the bottle. Stand back and watch the foam explode upward like giant toothpaste! The reaction happens fast — be ready!

Observe & Record

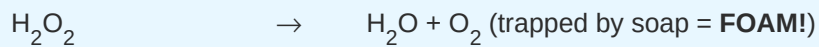
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Notice the colour, height, and texture of the foam. Is the bottle warm? This is called an **exothermic reaction** — it releases heat!

THE SCIENCE BEHIND IT

Why Does It Happen?

When yeast is added to hydrogen peroxide (H_2O_2), the yeast acts as a **catalyst** — a substance that speeds up a chemical reaction without being used up itself. The yeast rapidly breaks down the hydrogen peroxide into **water (H_2O)** and **oxygen gas (O_2)**. The dish soap traps all that oxygen, creating thousands of tiny bubbles that form the giant foam column.



Key Science Terms

- **Catalyst** — A substance that speeds up a reaction without being consumed.
- **Decomposition** — Breaking one substance down into simpler substances.
- **Exothermic** — A reaction that releases energy (heat) as it proceeds.
- **Oxygen** — The colourless gas released here that inflates the foam bubbles.

DISCUSSION QUESTIONS

Talk About It!

- Q1. What do you think would happen if we used more yeast?
- Q2. Why did we add dish soap? What would happen without it?
- Q3. Did you notice the bottle felt warm? Why do you think that happened?
- Q4. Can you think of other reactions in real life that produce bubbles or foam?
- Q5. Why is it important to wear goggles and gloves during experiments?

FUN ELEPHANT FACTS

Fun Facts About Elephants!

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| ■ | Elephants love to play in water and use their trunks to spray themselves cool. |
| ■ | An elephant's trunk has around 40,000 muscles — compared to about 640 in the entire human body! |
| ■ | Elephants are incredibly strong but are known to be gentle and empathetic animals. |
| ■ | Elephants are one of the few large animals that cannot jump! |

- They use their trunks like giant straws to drink, pick up objects, and communicate.

CLEAN-UP INSTRUCTIONS

How to Clean Up

- The foam is safe to touch once fully expanded — it is just soap and water.
- Rinse the bottle and tray thoroughly with water.
- Dispose of the foam down the drain with plenty of water.
- Ask an adult to safely dispose of any leftover hydrogen peroxide.
- Remove and wash gloves; rinse goggles with water.

BONUS IDEAS

More Science Fun to Try!

■ Baking Soda Volcano	Add baking soda to a model volcano, then pour vinegar in. Watch the fizzy lava flow!
■ Colour Mixing Magic	Fill clear cups with red, blue, and yellow water. Use pipettes to blend primary colours.
■ Floating & Sinking	Drop different objects into a bowl of water. Predict before each drop — will it float or sink?